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STOCHASTIC NETWORK PROGRAMMING FOR FINANCIAL PLANNING PROBLEMS*

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Several financial planning problems are posed as dynamic generalized network models with stochastic parameters. Examples include: asset allocation for portfolio selection, international cash management, and programmed-trading arbitrage. Despite the large size of the resulting stochastic programs, the network structure can be exploited within the solution strategy giving rise to efficient implementations. Empirical results are presented indicating the benefits of the stochastic network approach for the asset allocation case.

(FINANCIAL PLANNING; STOCHASTIC PROGRAMMING; GENERALIZED NETWORKS; SCENARIO ANALYSIS)

1. Introduction

Financial planning problems typically concern the (re)positioning of funds to achieve specific goals. In allocating financial resources, the available options must be analyzed from various perspectives. First, the potential earnings of alternative fund positions must be evaluated in conjunction with the costs associated with the transfers of funds. Second, consumption needs must be met and anticipated future deposits and liabilities must be properly taken into account. A multiperiod model becomes necessary.

Consideration of uncertainties is critical in financial planning. Major uncertainties include the returns of investment instruments, future borrowing rates and external deposit/withdrawal streams. Investors often seek to develop long-term strategies that hedge against uncertainties. Although imminent decisions are of primary importance, the strategies sought should also provide recourse decisions for subsequent periods under various contingencies.

Dynamic problems under uncertainty can be modeled as multistage stochastic programs, thereby capturing the dynamic aspects of the problem and allowing for a number of realizations of the uncertain quantities. Classic references on stochastic programs with recourse include Dantzig and Madansky (1961) and Walkup and Wets (1966). Stochastic linear programs with recourse have been applied to various problems including natural resources management (Somlyódy and Wets 1988, Gassmann 1987, Pereira and Pinto 1991), facility location (Wallace 1989), asset and liability management (Kusy and Ziemba 1986), and resource acquisition (Bienstock and Shapiro 1988).

Multistage stochastic programs exhibit a multiplicative growth in size with the number of decision stages and the number of realizations admitted at each stage. Dempster (1988), Ermoliev and Wets (1988, Chapter 1), and Varaiya and Wets (1988) address the complexity of dynamic stochastic programs. Practical applications have generally been restricted to two-stage models permitting uncertainty only in the cost coefficients and in the right-hand side vector of the constraints.

In this paper we develop stochastic generalized network programs for financial planning problems under uncertainty. Nonlinear networks have proved to be useful tools in an-

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alyzing a wide range of real-world problems (see Dembo et al. 1989). They possess many desirable properties, not least among them the existence of efficient solution methods that capitalize on the graph structure of the problem (e.g., Ahlfeld et al. 1987, Zenios and Lasken 1988). Certain stochastic formulations of networks are amenable to solution by decomposition methods (Mulvey and Vladimirou 1991a, b).

First, we develop a prototype dynamic network model that captures essential features of various discrete time financial decision problems. We allow for uncertainty in asset returns, interest rates and external deposits and withdrawals over time. The uncertain environment is represented by a set of discrete realizations (scenarios) of the uncertain quantities. We then show how the general model can be tailored to specific financial planning problems.

Discrete scenario models for related financial decision problems have been proposed in the past. Bradley and Crane (1972) and Lane and Hutchinson (1980) suggested dynamic linear programs for managing bond portfolios. Both studies directly solved the deterministic equivalent programs. Consequently, these models were restricted to the analysis of strategic uncertainties because of the rapid growth in the size of the deterministic equivalent with the number of postulated scenarios. Kusy and Ziemba (1986) employed multiscenario linear programs with simple recourse to model asset and liability management problems in banking. Only external deposits and withdrawals were explicitly treated as uncertain; other uncertain parameters were substituted by their expected values.

Simple recourse models allow uncertainty only in the right-hand side vector of the constraints and proportionately penalize constraint violations under the different scenarios. However, the impact of such constraint violations on the decisions at later periods cannot be assessed, thus limiting the dynamic features of the model. Stochastic programs with simple recourse can be solved efficiently by specialized methods (e.g., Wets 1983, Nazareth and Wets 1988, Qi 1986). Stochastic linear programs with general recourse matrices are typically solved by cutting plane algorithms derived from the L-shaped decomposition method of Van Slyke and Wets (1969). Computational shortcuts in these methods depend on a fixed (scenario invariant) recourse matrix (see, for example, Wets 1988).

Zenios (1991a) and Hiller and Eckstein (1991) recently proposed multiperiod stochastic programs for selecting portfolios of fixed-income securities. The model in Hiller and Eckstein (1991) constructs a static portfolio over a multiperiod horizon and ignores rebalancing opportunities in future periods, thus limiting the dynamic aspects of the portfolio planning problem. The model in Zenios (1991a) falls in the framework of the multiscenario generalized networks discussed in this paper.

In our multiscenario generalized network models uncertainty in asset returns directly affects the constraint coefficient matrix. Because of the absence of a fixed recourse matrix and the nonlinearity of the objective function we opted to solve the stochastic optimization programs with the progressive hedging algorithm of Rockafellar and Wets (1991). This procedure decomposes the problem to its constituent scenario subproblems, thus maintaining the network structure. For details on the adaptation of the progressive hedging algorithm to stochastic generalized networks refer to Vladimirou (1990). Decomposition demonstrates potential in coping with the large size of stochastic programs by partitioning the problem to manageable components. Exploitation of the network structure and the use of parallel computing yield significant additional benefits (Vladimirou and Mulvey 1991).

The stochastic network framework presented here encompasses a broad range of financial planning problems; it allows uncertainty in all important parameters and addresses the dynamic aspects of the decision problems including investment opportunities in futures/forward markets. Also, the solution approach represents an implementation of a new decomposition algorithm.

2. A Multiscenario Generalized Network Model

In this section we formulate a multiscenario generalized network program (MSN). The general model can be tailored to problems in portfolio management, currency exchange, trading arbitrage, and cash flow management. The common theme in these problems is the positioning of funds in various asset categories over several periods. Particularly challenging are dynamic cases involving decisions at multiple time periods, with information concerning uncertain parameters being progressively revealed during the planning horizon.

Let $\mathcal{J}$ indicate the set of asset categories and $\tau = 0, \dots, T$ the decision periods in the planning horizon; $\tau = 0$ is used to index initial parameters (deterministic) and imminent decisions. For each asset category $i \in \mathcal{J}$ we define the set $\mathcal{J}_i \equiv \mathcal{J} \setminus \{i\}$. A generalized network model is constructed by associating a network node with each asset-period pair and an arc with each transaction decision, as shown in Figure 1. Each arc may have an associated multiplier that augments the flow on the arc; the flow entering the arc at the tail node is scaled by the multiplier before reaching the head node. Thus, exchange, return, and borrowing rates are modeled by means of arc multipliers in the generalized network framework. External deposits/withdrawals correspond to supplies and demands at the nodes of the network. Accounting constraints that enforce a balance in the sources and uses of funds reduce to flow conservation equations at the nodes of the network.

Uncertainty is represented by a number of distinct realizations. Each complete realization of all uncertain parameters gives rise to a *scenario*; we denote by S the discrete set of all scenarios. Let $\pi_s > 0$ be the probability of occurrence for scenario $s \in S$; $\sum_{s \in S} \pi_s = 1$. Several scenarios may reveal identical values for the uncertain quantities up to a certain period—i.e., they share common information history up to that period. In order to avoid dependence on hindsight, scenarios that share common information history up to a specific period must yield the same decisions up to that period. Hence, the manner in which information progressively becomes available should affect the decision process. We will address the representation of the information structure following a description of the network model. We index parameters and decision variables by $s \in S$ to indicate their corresponding value under a particular scenario.

We denote by $b_i^\tau(s)$ the external inflow ($b_i^\tau(s) > 0$) or outflow ($b_i^\tau(s) < 0$) in asset category $i \in \mathcal{J}$ at period τ . The values of uncertain deposits/withdrawals for periods $\tau = 1, \dots, T$ must be revealed before or at time τ . Also, let $x_{ij}^{\tau t}(s)$ be the magnitude of a transaction decided at period τ to convert holdings in asset $i \in \mathcal{J}$ to asset $j \in \mathcal{J}_i$ at period t ($\tau \leq t \leq T$). For $t = \tau$ the transaction is carried out in the same period at which the decision is made (spot market). For $t > \tau$ the variables represent transactions in the forward or futures market; these decisions are made at period τ but the transactions are executed at time t . The multiplier on the arc that corresponds to transaction $x_{ij}^{\tau t}$ is denoted as $\zeta_{ij}^{\tau t}(s)$ and is assumed to be known at the time the transaction is decided (τ). This parameter reflects the forward exchange rate (if $\tau < t$) or a linear transaction cost (brokerage fee). For example, if the marginal transaction cost involved in such a transfer of assets under scenario $s \in S$ is $\kappa_{ij}^{\tau t}(s)$, then the corresponding arc multiplier is set to $\zeta_{ij}^{\tau t}(s) = (1 - \kappa_{ij}^{\tau t}(s))$. The flows of funds due to transfer decisions are depicted in Figure 1. Only a subset of the decision variables (arcs) is included in this figure to illustrate the basic structure of the network model.

Holdings in a particular asset class can be placed in interest-bearing instruments. Let $y_i^{\tau t}(s)$ be the amount of asset $i \in \mathcal{J}$ placed at period τ into an instrument maturing at the beginning of period t ($\tau < t$). The cumulative return $\rho_i^{\tau t}(s)$ on such an instrument may be uncertain at the time the decision is made (τ), but becomes known before, or at, maturity (t)—or by the end of the planning horizon (T) if $t > T$. For a multiperiod

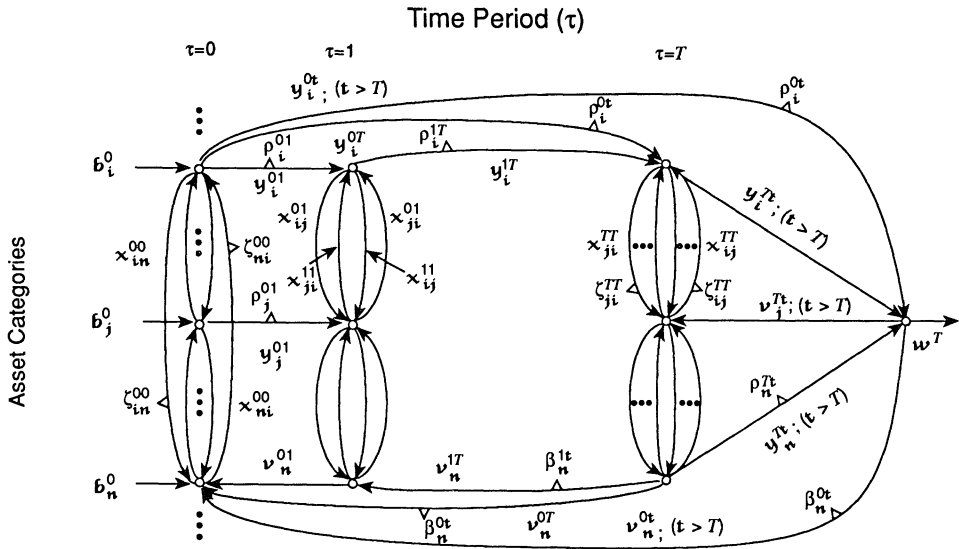


FIGURE 1. Basic Structure of the Network Model for Multiperiod Financial Planning Problems.

investment, we assume that cash spinoffs are reinvested in the same instrument; thus, full payment of principal and accrued interest takes place at maturity, akin to zero-coupon investments. Removal of this assumption would require nonnetwork side constraints; this is important in fixed-income securities whose cash payments depend on interest rate contingencies, like the mortgage-backed security model in Zenios (1991a). The effect of the investment decisions described above on the asset positions over the planning horizon is depicted in Figure 1.

A final collection node is used to gather the terminal net wealth—denoted by the variable $w_T(s)$ (see Figure 1). Investments with maturities extending beyond the end of the planning horizon “flow” into the collection node. The arc multipliers associated with such decisions are adjusted to reflect the market value of the respective asset at the end of the planning horizon—including accumulated returns up to that time. Thus, for an investment of asset $i \in \mathcal{I}$ at period τ in an instrument that matures at period t ($0 \leq \tau \leq T < t$) the corresponding arc multiplier becomes $(\rho_i^{\tau t}(s))^{(T-\tau)/(t-\tau)}$.

To include borrowing or short sales we define variables $v_i^{\tau t}(s)$ to represent the amount of funds in asset $i \in \mathcal{I}$ transferred from period t to provide leverage at period τ ($\tau < t$). The amount of asset $i \in \mathcal{I}$ that becomes available at time τ (i.e., loan principal) as a result of such a leverage decision is

$$\beta_i^{\tau_l}(s) \cdot v_i^{\tau_l}(s), \quad \beta_i^{\tau_l}(s) = 1/(1 + \delta_i^{\tau_l}(s))^{(l-\tau)} \leq 1,$$

where $\delta_i^{\tau t}(s)$ is the interest rate on a loan in asset $i \in \mathcal{I}$ at period τ with maturity at period t under scenario $s \in S$. $\beta_i^{\tau t}(s)$ (or $\delta_i^{\tau t}(s)$) is assumed to be known at the time the decision is made (τ). So, $v_i^{\tau t}(s)$ is the amount of asset $i \in \mathcal{I}$ (including accrued interest) that is needed at period t to repay the respective leverage decision made at period τ . Again, outstanding debts are paid back (as a lump-sum) at maturity. Refer to Figure 1 for the flow of funds due to leverage decisions.

Loans maturing after the end of the planning horizon are permitted; an arc representing such a decision originates at the collection node. The corresponding arc multiplier is adjusted, so that $v_i^{\tau t}(s)$ (for $0 \leq \tau \leq T < t$) represents the magnitude of the respective

outstanding liability at the end of the horizon, including accrued interest up to that time. Hence, the corresponding arc multiplier is set to $(\beta_i^{\tau t}(s))^{(T-\tau)/(t-\tau)}$.

Finally, let $x(s)$, $y(s)$, $v(s)$, $w(s)$ be the respective vectors of the decisions described above under scenario $s \in S$ and denote by $z(s) = \{x(s), y(s), v(s), w(s)\}$ the vector of all decision variables under scenario $s \in S$. Let $F(z(s), s)$ be a concave, continuously differentiable objective function that may depend parametrically on the scenarios.

We mentioned earlier that relevant quantities are indexed by $s \in S$ to indicate their dependence on a particular scenario and we noted that groups of scenarios may reveal identical values for the uncertain parameters up to a certain period. We specify the information structure by a scenario tree, which we use to impose the logical requirement for independence from hindsight. This condition—known as *nonanticipativity*—dictates that scenarios with common information history up to a particular period must yield the same decisions up to that period. A sample scenario tree is shown in Figure 2.

The scenario tree has a depth equal to the length of the planning horizon (T)—although this is not necessary. A stage in the scenario tree corresponds to a time at which new information becomes available; that is, a time when the values of some random variables are observed. Decision periods on the other hand represent the times at which rebalancing transactions take place. Multiple decision periods may be included within a single information stage. Such an aggregation may be applied, for example, to reduce the number of information stages, so as to reduce the size of the stochastic program. For simplicity in notation, in this paper we assume that each decision period corresponds to a new stage of the scenario tree.

Each scenario is associated with a path from the root node to a distinct leaf node (such as the highlighted path in Figure 2). The nodes on such a path identify the joint realizations of the uncertain parameters at each stage. The corresponding leaf node k_T distinguishes the respective scenario. The total number of leaf nodes equals the cardinality of the scenario set S . Paths through the same tree node k_τ at a certain stage τ correspond to scenarios that have identical realizations for the uncertain parameters up to that stage; these paths form a scenario subset $L(k_\tau) \subseteq S$ (see Figure 2). Hence, distinct tree nodes at the same stage partition the scenario set into disjoint subsets. The number of disjoint subsets at stage τ equals the number of tree nodes at that stage, which comprise the set $K(\tau)$.

By construction, scenarios that belong to the same subset $L(k_\tau) \subseteq S$ at stage τ have identical realizations for all uncertain parameters up to stage τ . Consequently, for arbitrary scenarios $s \neq s'$, if $s \in L(k_\tau)$ and $s' \in L(k_\tau)$, where $k_\tau \in K(\tau)$, $\tau = 0, \dots, T$, we have

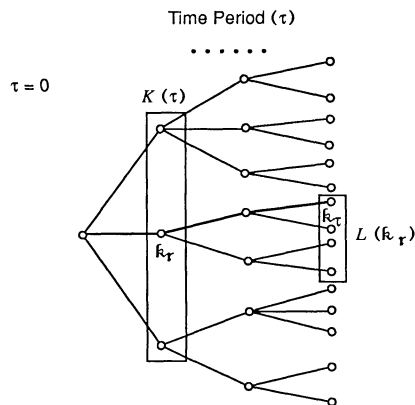


FIGURE 2. A Scenario Tree.

$$\begin{aligned}
b_i^t(s) &= b_i^t(s') & \forall i \in \mathcal{I}, & \quad 0 \leq t \leq \tau, \\
\zeta_{ij}^{t'}(s) &= \zeta_{ij}^{t'}(s') & \forall i \in \mathcal{I}, & \quad \forall j \in \mathcal{J}_i, \quad 0 \leq t \leq \tau, \quad t \leq t' \leq T, \\
\rho_i^{t'}(s) &= \rho_i^{t'}(s') & \forall i \in \mathcal{I}, & \quad 1 \leq t' \leq \tau, \quad 0 \leq t < t', \\
\beta_i^{t'}(s) &= \beta_i^{t'}(s') & \forall i \in \mathcal{I}, & \quad 0 \leq t \leq \tau, \quad 0 \leq t < t'.
\end{aligned}$$

We have defined a separate set of decision variables for each scenario. Hence, additional constraints are needed to enforce the logical requirement of nonanticipativity. The scenario tree is useful in this task. For every scenario subset at each stage we define additional variables corresponding to the decisions at the respective period. Hence, for each nonleaf node $k_\tau \in K(\tau)$, $\tau = 0, \dots, T-1$, of the scenario tree we define the following conformable vectors of artificial variables:

$$\begin{aligned}
X(k_\tau) &= \{X_{ij}^{\tau t}(k_\tau) : i \in \mathcal{I}, j \in \mathcal{J}_i, \tau \leq t \leq T\}, \\
Y(k_\tau) &= \{Y_i^{\tau t}(k_\tau) : i \in \mathcal{I}, \tau < t\}, \\
V(k_\tau) &= \{V_i^{\tau t}(k_\tau) : i \in \mathcal{I}, \tau < t\}.
\end{aligned} \tag{1}$$

We impose nonanticipativity by explicitly equating the conditional decisions of each scenario that belongs to a particular subset with the corresponding artificial variables for that subset. Therefore, we use the following constraints to ensure equality of decisions at a certain stage $\tau < T$ across all scenarios $s \in L(k_\tau)$ that share common realizations of uncertainty up to that stage (i.e., they belong to the same subset at that stage):

$$\begin{aligned}
x_{ij}^{\tau t}(s) - X_{ij}^{\tau t}(k_\tau) &= 0 \quad \forall i \in \mathcal{I}, \quad \forall j \in \mathcal{J}_i, \quad \tau = 0, \dots, T-1, \quad \tau \leq t \leq T, \\
&\quad \forall s \in L(k_\tau), \quad \forall k_\tau \in K(\tau),
\end{aligned} \tag{2}$$

$$\begin{aligned}
y_i^{\tau t}(s) - Y_i^{\tau t}(k_\tau) &= 0 \quad \forall i \in \mathcal{I}, \quad \tau = 0, \dots, T-1, \quad \tau < t, \\
&\quad \forall s \in L(k_\tau), \quad \forall k_\tau \in K(\tau),
\end{aligned} \tag{3}$$

$$\begin{aligned}
v_i^{\tau t}(s) - V_i^{\tau t}(k_\tau) &= 0 \quad \forall i \in \mathcal{I}, \quad \tau = 0, \dots, T-1, \quad \tau < t, \\
&\quad \forall s \in L(k_\tau), \quad \forall k_\tau \in K(\tau).
\end{aligned} \tag{4}$$

Nonanticipativity is trivially satisfied at the last stage (i.e. $\tau = T$) as each scenario leads to distinct decisions at the last stage.

We now state a general multiscenario formulation for discrete time financial planning problems under uncertainty:

$$[\text{MSN}] \quad \max \sum_{s \in S} \pi_s F(z(s), s) \tag{5}$$

subject to

$$\begin{aligned}
&\sum_{j \in \mathcal{J}_i} x_{ij}^{00}(s) - \sum_{j \in \mathcal{J}_i} \zeta_{ji}^{00}(s) x_{ji}^{00}(s) + \sum_{t>0} y_i^{0t}(s) - \sum_{t=1}^T \beta_i^{0t}(s) v_i^{0t}(s) \\
&\quad - \sum_{t>T} (\beta_i^{0t}(s))^{T/t} v_i^{0t}(s) = b_i^0(s) \quad \forall i \in \mathcal{I}, \quad \forall s \in S,
\end{aligned} \tag{6}$$

$$\begin{aligned}
& \sum_{t=0}^{\tau} \sum_{j \in \mathcal{J}_i} x_{ij}^{t\tau}(s) - \sum_{t=0}^{\tau} \sum_{j \in \mathcal{J}_i} \zeta_{ji}^{t\tau}(s) x_{ji}^{t\tau}(s) + \sum_{t > \tau} y_i^{t\tau}(s) \\
& - \sum_{t=0}^{\tau-1} \rho_i^{t\tau}(s) y_i^{t\tau}(s) + \sum_{t=0}^{\tau-1} v_i^{t\tau}(s) - \sum_{t=\tau+1}^T \beta_i^{t\tau}(s) v_i^{t\tau}(s) \\
& - \sum_{t > T} (\beta_i^{t\tau}(s))^{(T-\tau)/(t-\tau)} v_i^{t\tau}(s) = b_i^{\tau}(s) \quad \forall i \in \mathcal{J}, \quad \tau = 1, \dots, T, \quad \forall s \in S, \quad (7)
\end{aligned}$$

$$\begin{aligned}
& - \sum_{i \in \mathcal{J}} \sum_{\tau=0}^T \sum_{t > T} (\rho_i^{t\tau}(s))^{(T-\tau)/(t-\tau)} y_i^{t\tau}(s) \\
& + \sum_{i \in \mathcal{J}} \sum_{\tau=0}^T \sum_{t > T} v_i^{t\tau}(s) + w^T(s) = 0 \quad \forall s \in S, \quad (8)
\end{aligned}$$

$$z(s) \in \mathcal{Z} \quad \forall s \in S, \quad (9)$$

and the nonanticipativity constraints (2)–(4).

The equations in (6) specify that immediate decisions are governed by the initial holdings in each asset class and the existing market rates. Note that the initial asset holdings $b_i^0(s)$ are deterministic (constant across all scenarios). The current exchange rates $\zeta_{ij}^{00}(s)$ and borrowing rates $\beta_i^{0t}(s)$ are also deterministic. The proposed formulation, however, duplicates the constraints in (6) for each scenario because separate decision variables are defined for each scenario. This “split variable” approach facilitates the decomposition of the constraints into a separate set for each scenario.

The constraints in (7) balance the sources and uses of funds for each asset class at each time period. The definitional constraints in (8) specify the terminal net wealth $w^T(s)$ under each scenario $s \in S$. Program [MSN] includes investment instruments and loans with maturities beyond the end of the horizon. The multipliers associated with such decision variables in equations (6)–(8) are adjusted as described earlier, so that the corresponding inflows (outflows) at the collection node reflect the market value of the respective assets (liabilities) at the end of the planning horizon. The set $\mathcal{Z}$ in (9) represents nonnegative lower and upper bounds on the decision variables; bounds are imposed to control exposure to any particular asset category and to limit borrowing.

The constraints (6)–(9) decompose across scenarios. Disjoint scenarios are coupled only by the nonanticipativity constraints (2)–(4). Decision variables satisfying equations (6)–(9) can be described by arc flows on a generalized network graph. In fact, for each scenario the constraints (6)–(9) represent flow conservation conditions at the nodes of the generalized network shown in Figure 1. The topology of the network graph is the same for all scenarios; only the arc multipliers and supply/demand values differ between scenarios.

Inevitably, the size of program [MSN] grows rapidly with the number of decision stages and with the number of scenarios. Decomposition affords the opportunity to solve a number of smaller size subproblems. Preserving the network structure is equally important in order to use efficient, specialized algorithms. In the split-variable formulation the nonanticipativity constraints pose the only obstacle to decomposition.

The progressive hedging algorithm of Rockafellar and Wets (1991) (see also Wets 1989) operates on the split variable form of the problem as stated in [MSN]. The algorithm achieves the desired decomposition by dualizing the nonanticipativity constraints (2)–(4) which it incorporates by means of multiplier and (quadratic) penalty terms in the objective function, in the context of an augmented Lagrangian. A proximal point method is in turn applied to the resulting optimization program to resolve the nonseparability

of the quadratic penalty term. Thereby, program [MSN] is decomposed to its constituent scenario subproblems (generalized networks) that are solved independently by specialized codes. The individual subproblem solutions generally violate the relaxed nonanticipativity constraints. Consequently, the multipliers and the proximal term in the quadratic penalty are iteratively updated to progressively enforce the nonanticipativity constraints.

We now address the issue of choosing appropriate objective functions. The maximization of expected utility is a standard paradigm in financial analysis. When maximizing the expected utility of terminal net wealth the objective function becomes:

$$\max \sum_{s \in S} \pi_s \mathcal{U}(w^T(s)/q(s)). \quad (10)$$

$\mathcal{U}(w^T(s)/q(s))$ is a concave utility function of net wealth growth by the end of the horizon under scenario $s \in S$; $q(s)$ is the respective net present value of the external deposit/withdrawal stream:

$$q(s) = \sum_{\tau=0}^T (1 + \alpha)^{-\tau} \sum_{i \in \mathcal{I}} b_i^\tau(s), \quad 0 \leq \alpha < 1, \quad (11)$$

where α is a discount factor (normally the rate of return of a riskless asset).

Maximizing the expected utility of terminal net wealth is appropriate when the maximization of long-term utility is the main priority. Other investment goals, however, might place different preferences on wealth levels at various time periods in the planning horizon. Meyer (1976) suggests the use of discounted expected utilities for dynamic problems involving time preferences in the outcomes. Discounting expected utilities of net wealth at the discrete decision times yields the following objective:

$$\max \sum_{s \in S} \pi_s \sum_{\tau=0}^T (1 + \alpha)^{-\tau} U(w^\tau(s)). \quad (12)$$

$U(w^\tau(s))$ is a concave utility function of effective net wealth at period τ under scenario $s \in S$. The effective net wealth $w^\tau(s)$ at period τ is computed from the market value of total assets and liabilities held at that period. The value of assets/liabilities that mature after period τ are computed on the basis of accrued return/interest up to period τ :

$$w^0(s) = \sum_{i \in \mathcal{I}} b_i^0(s), \quad s \in S, \quad (13)$$

$$\begin{aligned} w^\tau(s) = & \sum_{i \in \mathcal{I}} \sum_{t=0}^{\tau} \sum_{t' > \tau} (\rho_i^{t'}(s))^{(\tau-t)/(t'-t)} y_i^{t'}(s) + \sum_{i \in \mathcal{I}} b_i^\tau(s) \\ & - \sum_{i \in \mathcal{I}} \sum_{t=0}^{\tau} \sum_{t' > \tau} (\beta_i^{t'}(s))^{(t'-\tau)/(t'-t)} v_i^{t'}(s), \quad \tau = 1, \dots, T, s \in S. \end{aligned} \quad (14)$$

The selection of a discount factor and an appropriate utility function should reflect the decision-maker's preferences. The objective obtained by substituting (13)–(14) in (12) is a sum of concave, nonseparable functions of the conditional decision variables for each scenario. Still, decomposition across scenarios is maintained.

The stochastic network model has several appealing features. Program [MSN] admits a multitude of objective functions to capture various decision goals and risk tolerance attitudes. Multiperiod investments and liabilities are included in an integrated framework. Trading activities are synchronized and balanced with internal earnings, liability obligations and external deposit/withdrawal streams to meet budget requirements over time under the postulated economic scenarios. The inclusion of forward transactions in the

opportunity set offers an additional advantage. Also, the generalized network model properly accounts for linear transaction costs by means of arc multipliers. The effect of transaction costs on budget constraints is directly considered in the flow conservation equations. In multiperiod models ($T > 1$) the impact of transaction costs can be amortized over multiple periods. Hence, high cost transactions can be selected if the potential earnings over multiple periods outweigh the transaction cost. Myopic models (i.e., $T = 1$) can not diffuse transaction costs over multiple periods.

Our model falls in the class of multistage stochastic nonlinear programs with recourse. Uncertainty is allowed in all important parameters. Fluctuating asset returns and interest rates over time are represented by scenario dependent arc multipliers. As a result, there is not a fixed recourse matrix since the arc multipliers directly affect the constraint coefficients. The model's effectiveness hinges on the ability to capture the uncertain environment by a moderate number of scenarios. Limiting the number of scenarios is important in order to maintain computational tractability. No distributional assumptions are made, however, allowing the use of alternative scenario generation methods. Time series and other econometric models, as well as sampling from assumed or empirical distributions, are some of the candidate scenario generation options.

Program [MSN] is based on a "split variable" representation that facilitates decomposition of the constraints and preserves the network structure in the scenario subproblems. We employ the progressive hedging algorithm and use the optimizer GENOS (Mulvey and Zenios 1987) to solve the subproblems so as to exploit the nonlinear generalized network structure. Implementation issues and computational results under various internal tactics are addressed in Mulvey and Vladimirov (1991a, 1991b). Table 1 summarizes computational results for a number of stochastic network programs drawn from portfolio management problems.

The results in Table 1 were obtained with a FORTRAN implementation of the progressive hedging algorithm executed on a Silicon Graphics IRIS 4D/70 workstation. This

TABLE 1
Summary of Computational Results for the Progressive Hedging Algorithm

PROBLEM	Assets	Periods	Scenarios	Problem Size				Solution Time (sec)
				Network Subproblems		Overall Problem		
				Nodes	Arcs	Rows	Variables	
Linear Programs								
LP.0	15	8	18	121	334	1923	5468	37.5
LP.4	15	4	70	61	163	3235	9133	321.1
LP.5	15	8	48	121	335	5103	14529	181.3
LP.2	15	6	80	91	248	6095	17313	665.7
LP.3	15	8	72	121	335	7647	21777	293.6
Nonlinear Programs								
K.NLP	10	4	41	45	116	1405	3876	405.2
N.NLP	10	6	39	67	178	2195	6106	365.7
NLP.4	15	4	70	61	163	3235	9133	189.7
NLP.8	15	8	36	121	335	3831	10905	630.1
NLP.2	15	6	80	91	248	6095	17313	924.9
NLP.3	15	8	72	121	335	7647	21777	1292.8
NLP.9	15	8	100	121	335	10615	30233	1874.6

computer has a RISC based architecture (80 ns clock speed and 8 Mbytes of RAM) and runs Unix V3.1. The f77 compiler was used with the default optimization level.

A more compact formulation of the multiscenario program is possible. In this approach, a separate set of decision variables at a certain stage is defined only for each distinct subset of scenarios at that stage—akin to the artificial variables described in (1). Thus, nonanticipativity is implicitly considered in the definition of the conditional decision variables. The compact program form can be obtained, for example, by substituting the scenario dependent variables in (6)–(9) with the respective artificial variables to which they are equated in (2)–(4), and then eliminating all replicated constraints. Indeed, this is the approach commonly used in formulating multistage stochastic programs. The reduction in the size of the overall program is obtained with a sacrifice of the network structure and the decomposable form of the constraint set. The size of the multiscenario program in its compact form is reported in Table 1 under the heading “overall problem.”

The test problems in Table 1 do not reflect the size of problems (in the number of assets and scenarios) that would be encountered in many practical settings, but they serve as instructive illustrations. Dynamic stochastic programs remain among the most computationally demanding problems in numerical optimization, yet recent algorithmic developments and advancements in computing technology now make feasible the application of stochastic programs to practical financial planning problems. Parallel implementations of the progressive hedging algorithm on various multiprocessor systems achieved near linear speedups (Vladimirou and Mulvey 1991). Recently, Nielsen and Zenios (1991) demonstrated that strictly convex stochastic network problems can be solved very efficiently with appropriate massively parallel row-action methods. They were able to solve stochastic networks with thousands of scenarios with specialized implementations of row-action algorithms on the massively parallel Connection Machine CM-2. These studies showed that parallel computing and the exploitation of the network structure provide a means for solving large-scale stochastic network problems.

3. Financial Planning Applications of Dynamic Networks

Network models had been proposed for deterministic financial planning problems in the past (eg. Christofides et al. 1979, Crum and Dye 1981). These models did not consider the impact of uncertainty on financial positions and could not be used to develop adaptive long-term hedging strategies. Dembo et al. (1989) discuss a mean/variance portfolio selection model with network structure. Mean/variance models are based on the assumption that either the asset returns follow a multivariate normal distribution, or that the decision-maker's preferences are described by a quadratic utility function (see, for example, Elton and Gruber 1981). Moreover, mean/variance models are restricted to a single period horizon and, thus do not directly address multiperiod asset and liability management problems. For recent work in the area of financial optimization see the volume edited by Zenios (1991b).

The stochastic generalized network framework [MSN] developed in this paper accounts for the dynamic aspects of financial planning problems (including opportunities in the futures and forward markets), considers uncertainty in all relevant parameters, accommodates arbitrary discrete distributions for the random parameters by means of a moderate number of scenarios (so as to capture covariational elements), and admits different utility functions to represent the decision-maker's preferences and risk tolerance. This section illustrates how several prominent financial problems can be represented as special instances of the multiscenario network framework [MSN].

The first example concerns the problem of managing investment portfolios. The multiscenario network incorporates assets and liabilities in an integrated approach defined

as the *surplus portfolio model* (Mulvey 1988). Next, we show that various popular strategies for financial arbitrage can be modeled as multiscenario networks, including covered interest arbitrage and programmed trading. The final example takes up the problem of international cash management. Herein, a multinational corporation is faced with the task of managing its short term assets, primarily cash. Many complex strategies are possible, such as internal hedging, futures/forward market hedging, and option-based strategies. Again, multiscenario network models can address important instances of this problem.

3.1. Asset Allocation—Portfolio Management

Asset allocation concerns the distribution of funds to aggregate asset categories, while active portfolio management involves revisions of investment portfolios over time. Both problems are of interest to pension plan managers and large institutional investors. Portfolio selection under uncertainty has been extensively studied; see, for example, Markowitz (1959), Rubinstein (1975), Perold (1984), Levy and Sarnat (1984), and their references. Previous studies, particularly those based on the mean/variance paradigm of Markowitz (1959), covered single period models.

We now adapt the [MSN] model to address the dynamic aspects of multiperiod portfolio management problems. The returns of various investment instruments constitute the major element of uncertainty; again, they are modeled as scenario dependent arc multipliers. The resulting network model for each scenario has the basic structure shown in Figure 3.

Instead of direct transactions between each pair of assets, all sale ($x_{ic}^\tau(s)$) or purchase ($x_{ci}^\tau(s)$) transactions are coordinated through a cash node at each period τ (see Figure 3). Recall that variables are indexed by $s \in S$ to indicate their dependence on a particular scenario. Linear sale ($\kappa_{ic}^\tau(s)$) or purchase ($\kappa_{ci}^\tau(s)$) transaction costs are absorbed by means of multipliers ($\zeta_{ic}^\tau(s) = 1 - \kappa_{ic}^\tau(s)$, $\zeta_{ci}^\tau(s) = 1 - \kappa_{ci}^\tau(s)$) on the respective arcs that represent the portfolio rebalancing decisions at the corresponding period τ . Asset holdings in the revised portfolio are carried forward to the next period (variables $y_i^\tau(s)$ in Figure 3). The returns on these single-period investments are represented by scenario dependent

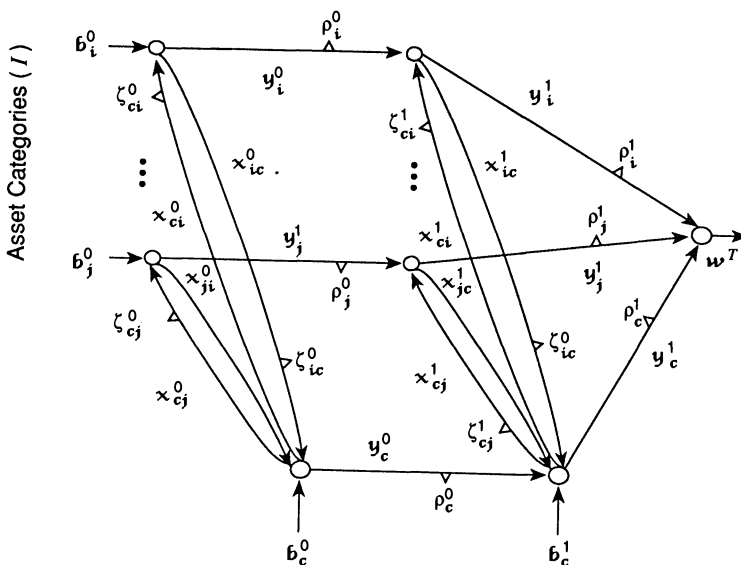


FIGURE 3. Basic Network Structure of the Multiperiod Portfolio Revision Model.

arc multipliers $\rho_i^T(s)$ to account for uncertainty. In this example we use only investments with single-period maturities; transactions in the forward market are also excluded.

Portfolio revisions take place at the discrete time periods in the planning horizon. The transaction costs are deducted at the times of portfolio revisions, since they appear in the form of arc multipliers in the flow balance equations of each period. Although Figure 3 illustrates only a two period model, the same basic network structure is repeated for all subsequent periods. At the end of the horizon all asset holdings “flow” into a collection node that gathers the terminal net wealth $w^T(s)$ under the respective scenario. Short sales and borrowing can be included by backward arcs as described in the previous section. Deposits and liabilities appear as supplies ($b_c^T(s) > 0$) and demands ($b_c^T(s) < 0$) at the cash nodes, respectively.

The objective function is based on expected utility maximization. The selection of a utility function that accurately represents the decision-maker’s risk tolerance attitude poses a modeling choice. However, certain classes of utility functions have emerged as the norms in practice; prominent among them is the family of *isoelastic utilities* (Hakansson 1971, 1974). Therefore, we substitute for the utility of terminal net wealth of equation (10) the isoelastic function:

$$\mathcal{U}(w^T(s)/q(s)) = \frac{1}{\gamma} \left(\frac{w^T(s)}{q(s)} \right)^\gamma, \quad \gamma \leq 1. \quad (15)$$

γ is a risk tolerance parameter and $q(s)$ is the net present value of the external deposit/withdrawal stream, as specified in (11).

Isoelastic utilities possess desirable properties: they are characterized by constant risk aversion and span a continuum of risk tolerance attitudes from *risk neutrality* ($\gamma = 1$) to *absolute risk aversion* ($\gamma \rightarrow -\infty$). The isoelastic family also includes the logarithmic utility (*growth optimal strategy*). Properties of logarithmic utility models have been extensively studied (eg. Rubinstein 1975); most important is the fact that they maximize the long-term expected growth rate of wealth. As a result, this *growth optimal policy* has often been used to derive dynamic investment strategies (see, for example, Taksar et al. 1988).

Hakansson (1971) proved that for isoelastic utilities, the solution of multiperiod investment problems reduces to the solutions of successive myopic problems under certain conditions. This result is based on the following assumptions: absence of transaction costs, perfect liquidity, no external deposits/withdrawals during the planning horizon, only single period investments, and intertemporal stochastic independence of asset returns. Hakansson (1974) also showed that under the same assumptions a broad class of utility functions converge to a member of the isoelastic family with a corresponding convergence of the investment policies generated by the respective myopic models.

Grauer and Hakansson (1982, 1985, 1986) reported encouraging results in simulations involving the successive application of myopic isoelastic utility models for asset allocation, using historical data. However, in their studies the first four of the above assumptions were artificially enforced. Their models considered only a budget constraint and margin requirements to control borrowing. The absence of transaction costs made asset holdings interchangeable and reduced the problem in each period to a simple portfolio construction. Our multisenario generalized networks account for linear transaction costs, making the opportunity set dependent on the composition of initial asset holdings. Multiperiod network models also consider dynamic effects and long-term investment goals.

Other utility functions can also be used. Kallberg and Ziemba (1983) reported that for investment problems with short planning horizons, models using utilities of different functional form but with similar risk aversion measures tend to produce similar investment

decisions. This provides the flexibility to select alternative utility functions if computational or other considerations dictate so.

We have tested myopic as well as multiperiod generalized network models for portfolio management in simulations using historical data. The simulations covered the period from the first quarter of 1979 to the first quarter of 1988—37 quarterly revisions in all. The myopic models had a planning horizon of one quarter, which allowed a single portfolio revision (i.e., $T = 1$). The multiperiod models had a planning horizon of one year and allowed portfolio revisions at each of the four quarters (i.e., $T = 4$). In all models the maximization of the expected utility of wealth at the end of their respective planning horizons was used as the objective function (cf. (10)). The utility functions were members of the isoelastic family (15).

We followed a rolling horizon approach in the simulations. At each quarter in the simulation, a multiscenario network—either a myopic or a 4-period model—was constructed and solved. The model's recommendations for the period immediately ahead were retained and implemented. For the myopic models this was the single portfolio revision in the planning horizon; for the 4-period models this was the portfolio revision recommended for the first of the four quarters in the year-long planning horizon. The individual asset returns observed during the period were applied to these investment policies, yielding the realized returns and determining the actual asset positions in the next quarter. A new model—myopic or 4-period—was constructed with the new initial portfolio composition and a new set of asset return scenarios, and the same procedure was repeated in the next quarter.

We initialized all simulations with a unit wealth in cash in the first quarter of 1979. We did not include liabilities or external cash flows; short sales and leverage decisions were also excluded in these examples. Clearly, these examples did not fully utilize all the features and modeling flexibilities allowed by the general model [MSN]. All internally generated revenues were reinvested in the following asset categories (comprising the set $\mathcal{J}$):

S&P 500	Standard and Poor 500 (large U.S. stocks);
SM CAP	smaller U.S. stocks;
CASH	3 month T-bills (riskless asset);
INT GOVT	intermediate term government bonds;
LT GOVT	long-term government bonds;
LT CORP	long-term corporate bonds;
HI YIELD	high yield securities;
EAFE	international stocks (Europe, Australia and Far East);
REIT	real estate;
VENT CAP	venture capital.

The scenario generation is an important step in the modeling phase. This task is complicated by the fact that asset returns are typically correlated; they are influenced by major economic factors and by within-market fluctuations. The distribution of uncertain asset returns is unknown. We have experimented with a simple probability approach advocated by Grauer and Hakansson (1982) for myopic models, whereby a number of the most recent observations of joint asset returns—assigned equal probability of occurrence—is used as the set of scenarios. In the rolling horizon context, we used the 24 most recent quarterly observations (6 years) for scenario generation purposes. These observations provided 24 scenarios for the asset returns in the single quarter horizon of the myopic models. In the multiperiod case, the joint asset returns observed for 4 consecutive quarter sequences in the 6-year calibration period were used as the scenarios, and were assigned equal probabilities. Hence, each 4-period model had a set of 21 scenarios. No forward information was used at any point in the scenario generation process.

We have also applied a scenario generation process based on principal component

analysis. In this approach, the eigenvectors of the covariance (or correlation) matrix of asset returns—obtained from past data—provide a number of uncorrelated random variables (principal components). Scenarios are then generated by sampling from the empirical or the derived distributions of the principal components. In the rolling horizon context, the covariance (or correlation) matrices were computed on the basis of the 24 most recent quarterly observations of the asset returns. Thus, the covariance matrix was reconstructed at each quarter in the simulation, an eigenvalue problem was solved, and a new set of asset return scenarios was obtained by sampling from the distributions of the principal components. For more details on the scenario generation process refer to Mulvey and Vladimirou (1989).

These simple procedures provided adequate representation of uncertainty as evidenced by the simulation results. Practical applications that involve many financial instruments will likely require consideration of a much larger scenario set. If the stochastic generalized network model is applied in portfolio management problems for fixed-income securities then a scenario generation process based on an arbitrage-free term structure model would be a more appropriate choice; such models are discussed in Zenios (1991a) and Hiller and Schaack (1992). Multiple scenario generation methods may have to be combined in applications that involve financial instruments with widely different risk-return characteristics and interest rate sensitivity, in order to generate a complete set of scenarios. Scenario generation methods are typically application specific. The development of proper scenario generation methods for various applications and effective scenario reduction techniques—needed to ensure computational tractability of stochastic programs—remain open research issues. These issues must be satisfactorily resolved for stochastic programming models to be widely embraced by the practicing community.

The performance of various portfolio management models was assessed in terms of the geometric mean and standard deviation of the realized (actual) portfolio returns over the simulation period (1979Q1–1988Q1). The standard deviation measures variability in achieved portfolio returns, while the geometric mean represents the achieved overall growth rate over the simulation period. These ex-post measures allow a comparison in the performance of different models; that is, models with different planning horizons and/or different objective functions. Geometric means and standard deviation values are reported in annualized units in the results that follow.

Several versions of multiscenario generalized network models were compared against each other, as well as against the performance of constant-mix portfolio strategies. Constant mix portfolios were maintained by redistributing asset holdings whenever fluctuations in asset returns disturbed the target composition of the portfolio. The compositions of some constant mix portfolios used in the analysis—given as fractions of total asset holdings in any given period—are shown in Table 2. These benchmark compositions were suggested by financial analysts as representative portfolio strategies.

The multiperiod generalized network models clearly outperformed the benchmark portfolios, as demonstrated in Figure 4. The 4-period network models also outperformed all individual asset categories, as well as respective myopic models with the same utility functions. Higher portfolio returns were produced by the dynamic network models at the same (or lower) level of variability (risk) in total returns. The resulting benefits reflect steeper and more stable growth paths. The relatively low variability in total returns exhibited by the benchmark strategies is attributed to the inclusion of low risk assets (primarily CASH) in these portfolios; note the increase in variability of portfolio returns for benchmarks BP5 and BP8. However, the 4-period network models also dominated more aggressive constant-mix strategies.

The performance of multiscenario network models tracked rather closely the expected growth paths—see Figure 5 for an example. The forecasted paths were determined by

TABLE 2
Compositions of Constant-Mix Benchmark Portfolios

Benchmark Portfolio	Proportion of Total Asset Holdings			
	S&P 500	LT GOVT	LT CORP	CASH
BP1		0.40	0.40	0.20
BP2	0.20	0.30	0.30	0.20
BP3	0.45	0.20	0.25	0.10
BP4	0.55	0.15	0.20	0.10
BP5	0.60	0.40		
BP6	0.65	0.10	0.15	0.10
BP7	0.75	0.15		0.10
BP8	0.90	0.05	0.05	
BP9	Equal proportions in the 10 asset classes			

applying the one-quarter expected portfolio return (found from the expected values of the individual asset returns, computed on the basis of the 24 most recent observations) to the actual wealth level at the beginning of the corresponding period. Thus, the network models generated active portfolio revision policies with good upside performance. The results indicate that dynamic network models in conjunction with simple scenario generation procedures can provide effective planning tools for complex investment problems.

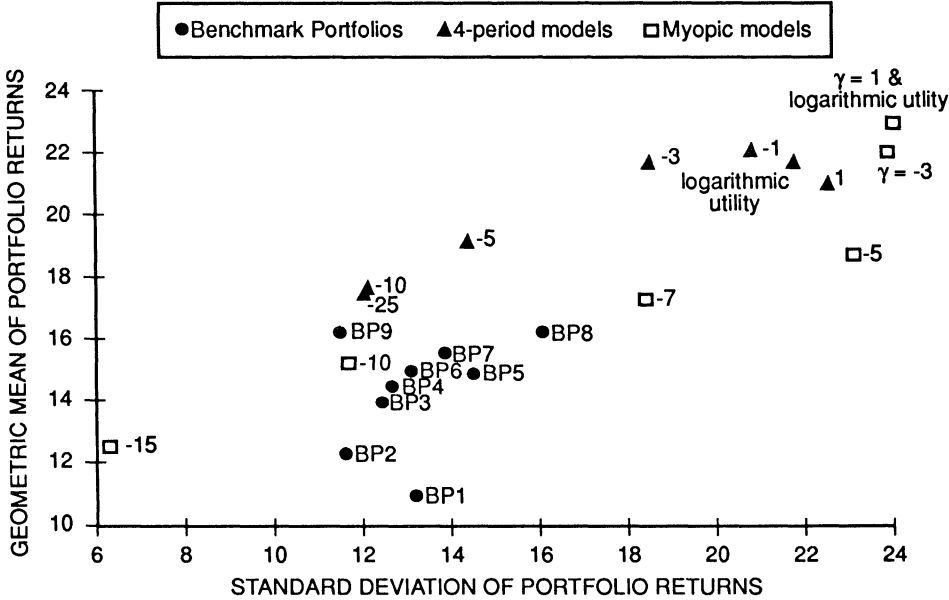


FIGURE 4. Performance Comparison for Portfolio Management Models; 1979Q1–1988Q1 Simulations (Quarterly Revisions, No Transactions Costs, No Bounds on Individual Asset Holdings).¹

¹ This figure compares the geometric means and standard deviations of cumulative portfolio returns observed over the entire simulation horizon (1979Q1–1988Q1) with alternative portfolio revision models. Results are presented for the constant-mix benchmark portfolios of Table 2, as well as for single-period and four-period stochastic network models. Isoelastic utility objectives were used in the stochastic network models; the risk tolerance parameter (γ) identifying the isoelastic utility used with the corresponding model is indicated for each data point in Figure 4.

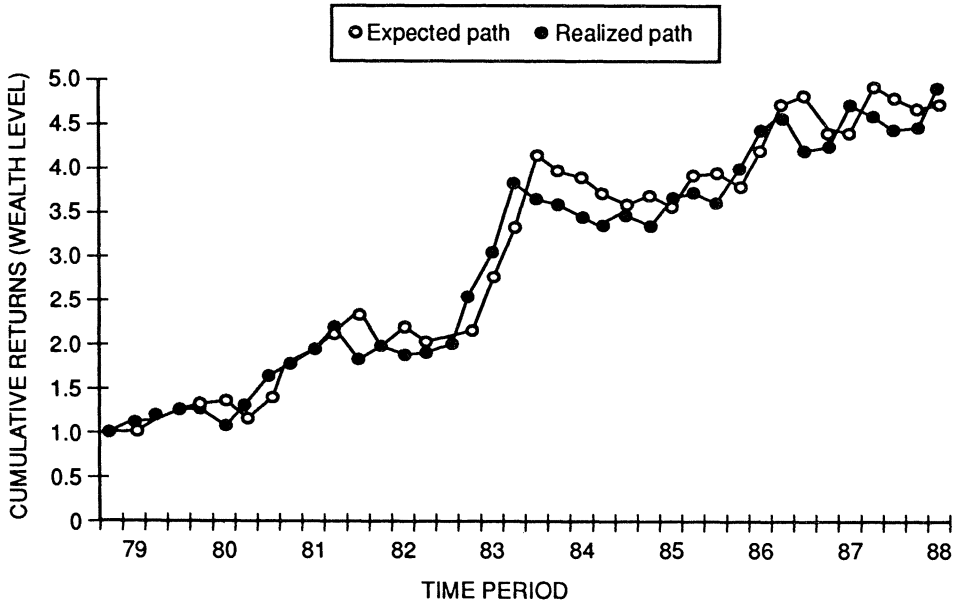


FIGURE 5. Actual and Forecasted Cumulative Growth Paths with 4-Period Model (Growth Optimal Utility, with Transaction Costs, No Bounds on Individual Asset Holdings).

Tests were run with and without transaction costs, as well as with and without upper bounds on individual asset holdings. Space does not permit a detailed discussion of all the strategies analyzed; additional results are presented in Mulvey and Vladimirov (1989). The marginal purchase and sale transaction costs were set to deterministic (scenario invariant) values at each decision period as follows: $\kappa_{ci}^\tau(s) = \kappa_{ic}^\tau(s)$, $i \in \mathcal{I}$, $\tau = 1, \dots, T$, $\forall s \in S$. These parameters determined the multipliers ζ_{ci}^τ and ζ_{ic}^τ on the arcs that represented the sale and purchase decisions in the network models as was discussed earlier. The following values—suggested by financial analysis—were used for the marginal transaction costs:

S&P 500	0.3%	SM CAP	1.0%	CASH	0.0%
INT GOVT	0.2%	LT GOVT	0.2%	LT CORP	0.2%
HI YIELD	0.4%	EAFE	2.0%	REIT	6.0%
VENT CAP	4.0%				

The 4-period models consistently outperformed their myopic counterparts, indicating that inclusion of dynamic features is essential in long-term investment planning problems. Particularly, multiperiod models exhibited a more prudent behavior when transaction costs were included, avoiding wide swings in portfolio compositions for small fluctuations in asset returns, and maintaining diversified portfolios. This behavior is attributed to the look-ahead capability of multiperiod models. The inclusion of bounds on individual asset holdings promoted further diversification.

Models using growth optimal (logarithmic) utility functions always came at, or near, the high end of the spectrum in terms of the achieved geometric mean values for total portfolio returns. Similarly, the 4-period models generally increased portfolio diversification for increasingly risk averse strategies (decreasing γ values), with a consequent reduction in the variance of the realized portfolio returns. Compare, for example, the portfolio compositions recommended by 4-period models with different isoelastic utilities

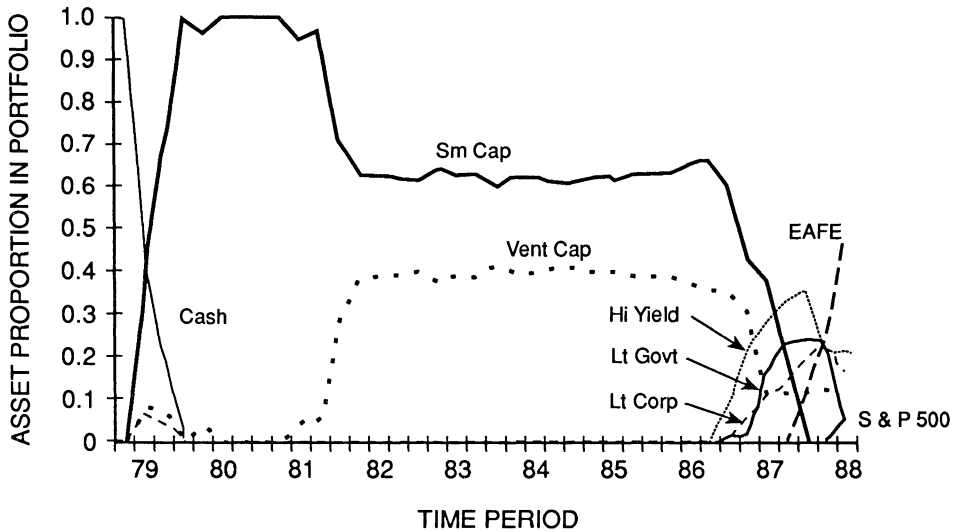


FIGURE 6. Portfolio Compositions Produced by 4-Period Model (Growth Optimal Utility, with Transaction Costs, No Bounds on Individual Asset Holdings).

for terminal net wealth that are reported in Figures 6 and 7. The observed behavior of the multisenario network models is consistent with expected utility theory.

Diversification was the preferred strategy for overcoming risk in multiperiod models as the value of the risk aversion tolerance γ was decreased. In myopic models, on the other hand, highly risk averse strategies ($\gamma \leq 0$) generally produced portfolios with substantial positions in riskless assets. Multiperiod models favored diversification even in the absence of transaction costs or bounds on individual asset holdings. Figure 6 shows the portfolio compositions over the simulation period recommended by the 4-period model using a logarithmic utility for terminal net wealth. Note particularly the increasingly diversified positions taken by the model starting in 1986. The selective diversification

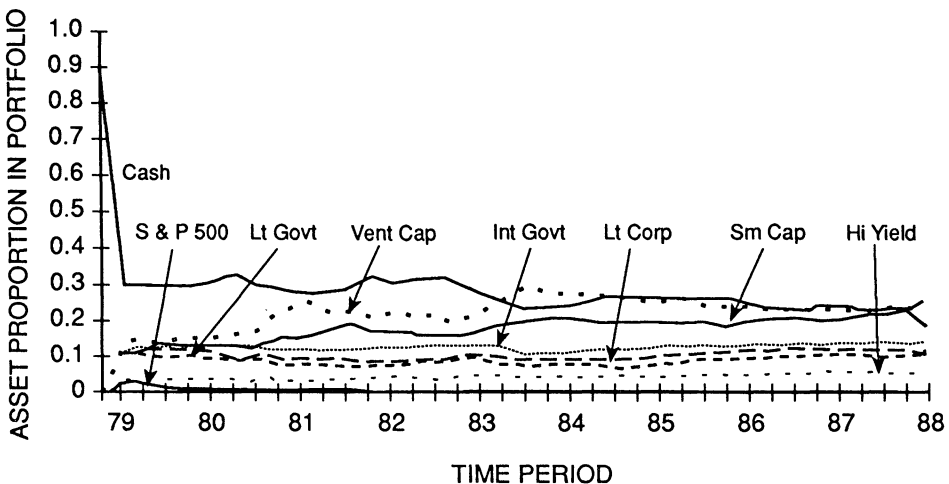


FIGURE 7. Portfolio Compositions Produced by 4-Period Model (Risk Averse Isoelastic Utility ($\gamma = -5$), with Transaction Costs, No Bounds on Individual Asset Holdings).

followed by multiperiod models dampened the effects of adverse investment conditions in the third quarter of 1987. These empirical results indicate that the use of simple scenario forecasts based on historical data is not without merit.

The network model bases its decisions on anticipated economic conditions over a multiperiod horizon. When maximizing monotonically increasing utility functions, the model increases exposure to risky assets only when their anticipated earning potential exceeds that of riskless assets. The converse occurs when the set of scenarios depicts a poor performance of the risky assets relative to riskless assets. Such behavior resembles strategies for purchasing portfolio insurance. Black and Perold (1990) and Perold and Sharp (1988) studied portfolio insurance strategies and showed that they provide convex payoff curves. Hence, the proposed model should provide effective downside protection as well as good upside potential. The effect of market reversals—in which constant-proportion portfolio insurance strategies do not perform particularly well (Black and Perold 1990)—can be considered with scenarios exhibiting oscillating asset returns over time.

3.2. Trading Arbitrage

Several popular examples of financial arbitrage fall within the [MSN] framework. For our purposes, arbitrage is defined as a riskless investment opportunity whereby money (or another commodity) is transferred through the network in the form of a loop, and the product of the multipliers for arcs on the loop is greater than unity. Consequently, a net gain is obtained by executing the sequence of transactions prescribed by the arcs on the loop. A simple example appears in Figure 8.

Here, three currencies—dollars, pounds sterling and Deutch marks—and their respective pairwise, spot exchange rates are displayed. A path around the loop of nodes $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ provides an arbitrage opportunity, as the product of the arc multipliers $0.611 \cdot 2.7740 \cdot 0.5910 = 1.003$. Transaction costs are included in the definition of the arc multipliers. Arbitrage cannot continue indefinitely; the exchange rates readjust as trades take place along the loop. Eventually equilibrium occurs when arbitrage opportunities disappear. Arbitrage opportunities reemerge, however, since currency movements are driven by outside economic forces—such as the actions of speculators and hedgers, as well as by Central Bank interventions. Upper bounds on the arcs are required to limit arbitrage transactions and to ensure boundedness of the optimization program.

Arbitrage becomes more complex and perhaps more likely to occur when trades at multiple time periods are considered. Figure 9 depicts an example of arbitrage involving transactions in the futures/forward market. Two assets are relevant here: gold and cash. In this case arbitrage requires that four simultaneous transactions take place: (1) gold is

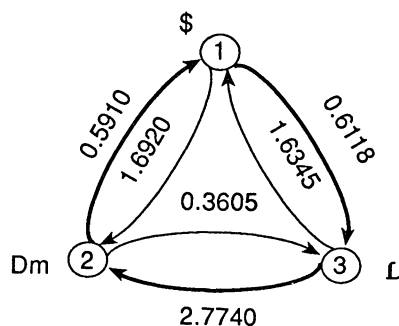


FIGURE 8. A Network Example of Spot Market Arbitrage.

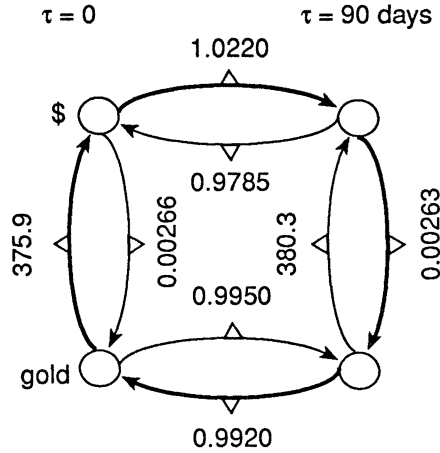


FIGURE 9. An Example of Arbitrage Opportunity Involving Forward Transactions.

borrowed; (2) gold is sold in the spot market; (3) cash is invested in an interest bearing instrument, and (4) gold is bought in the futures market. Again, the transactions are done without risk and persist until the market prices equilibrate. Trades of numerous currencies (or commodities) may be involved in such an arbitrage opportunity. In the case of currencies, the situation is called “covered-interest arbitrage.” More advanced strategies involve baskets of commodities (or stocks). The much analyzed programmed-trading also fits this structure. The unusual aspect of this strategy involves the simultaneous purchase or sale of many S&P500 stocks. Efficient computerized execution is critical in this situation.

The purpose of including arbitrage within the [MSN] context is to show that arbitrage strategies can be readily embedded in more general investment strategies without additional complexity to the model. The actual implementation depends upon the availability of real-time data and the ability to execute instantaneously multiple trades. The stochastic network program must be solved as quickly as possible.

3.3. *International Cash Management*

We now discuss how the [MSN] framework can be used to address the management of short term investments for a multinational corporation. This problem—called “international cash management”—is gaining in importance as markets become interconnected. Numerous US companies generate well over 50% of their business overseas. Currency fluctuations can be disruptive to a division’s profitability. A coordinated cash management system is thereby needed due to the complexity of the decisions. Optimization models for the deterministic case of this problem have been proposed by Rutenberg (1970) and Crum and Dye (1981). We believe that the [MSN] framework is well suited to address multiperiod cash management problems under uncertainty.

Since the investment decisions encompass multiple countries, the network must be expanded in an additional dimension to handle the currency aspects (see Figure 10). Investments must be considered within a country as well as across countries (currencies). Depending upon the number of currencies, and the level of detail in investment opportunities within a country, the network cash management model can become rather large. Therefore, the number of time periods must be kept relatively small.

The following elements of uncertainty need to be considered. First, supplies and demands for cash are possible at multiple locations at each time period. Anticipated cash

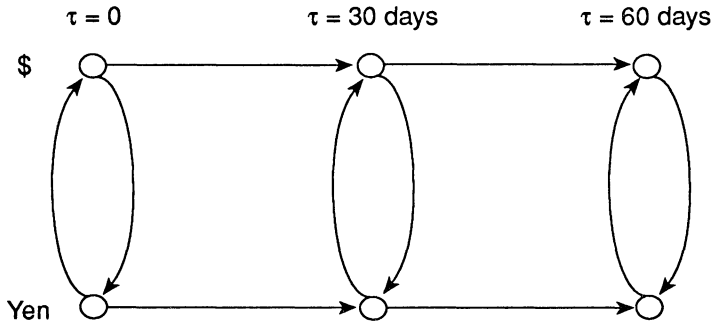


FIGURE 10. A Multiple Currency Cash Network.

supplies/demands depend upon the company's projected performance (or needs) at various locations, as a function of projected economic conditions. Revenues in a certain region may depend upon local economic factors, such as interest and inflation rates, or GNP. Thus, scenario dependent supplies/demands must be specified at the appropriate nodes of the network.

A second critical uncertainty entails the cross-exchange rates. The scenario generation process must postulate various economic conditions in order to provide a representative set of currency fluctuations. Again, the cross-exchange rates are represented by arc multipliers in the generalized network model.

In addition, futures/forward markets must be modeled. Risk averse companies will generally desire to hedge as much risk as possible. The commodity futures market provides an ideal vehicle. Again, the scenario generation process must develop exchange rates for forward transactions.

4. Conclusions and Future Work

This paper demonstrates that a variety of important financial planning problems can be represented in the framework of multiscenario generalized networks. Exploitation of the problem's structure and the use of parallel computing have led to long strides towards developing a capability to address large-size stochastic networks arising in practical settings.

The replication of the network structure under the different scenarios suggests the use of decomposition schemes that can be parallelized. Stochastic network problems can be efficiently solved with the progressive hedging algorithm by exploiting the network structure (Mulvey and Vladimirou 1991a, 1991b). Adaptations of the progressive hedging algorithm have achieved near linear speedups on various multiprocessor systems (Vladimirou and Mulvey 1991). Massively parallel row-actions methods for stochastic networks have proved especially promising (Nielsen and Zenios 1991).

Other solution alternatives include specializations of interior point algorithms to capitalize on the block-structured forms of multiscenario optimization programs. Progress has recently been made in this direction (see Mulvey and Ruszczyński 1992 and Lustig et al. 1991).

Inevitably the size of program [MSN] grows multiplicatively with the number of scenarios and the number of decision stages in the model. The scenario generation process is a critical aspect; appropriately refined scenario generation methods are needed for various problem classes. Statistical methods—such as importance sampling (Dantzig and Glynn 1990) and cluster analysis (Mulvey and Crowder 1979)—can limit the required number of scenarios to properly capture uncertainty and maintain computational tractability of the resulting stochastic programs. These issues are the subject of active research.

Besides limiting the number of scenarios, other approximation schemes often used in order to overcome the rapid increase in program size with the number of stages involve the truncation of long horizons and the aggregation of multiple periods into a single stage in stochastic formulations. The full effects of such approximation schemes on the solution of stochastic programs have not been fully studied. These important issues suggest promising research venues. Directions into the effect of horizon truncations in stochastic programs are given in Flåm and Wets (1985).

Still, it appears unlikely for the foreseeable future that all practical problems will be solvable, even by means of parallel computers. Improvements in computing capabilities—however significant or rapid—are consistently surpassed by the computational needs of increasingly large and more elaborate models for complex systems. Multistage stochastic programs will remain among the most computationally demanding problems in numerical optimization.

A promising avenue lies with the integration of stochastic optimization and multiperiod simulation. Such integrations can serve several purposes. First, simulation (or selective sampling) procedures can be valuable tools in the context of scenario generation. Further, we envision simulation being used as a means of assessing and validating alternative stochastic programming models. The effectiveness of alternative scenario reduction procedures or other aggregation and approximation techniques can be evaluated through multiperiod simulations in a rolling horizon fashion. The results of such simulations will measure the errors introduced by limiting the number of scenarios, or by aggregating multiple stages of stochastic programs, and will also help examine the stability of solutions in stochastic programs under different sets of scenarios. The simulation results also provide details about the distribution of outcomes over the planning horizon. Such information is valuable in calibrating particular models. For example, investors can get a sound idea of the nature of their problem by studying the simulation results, in order to choose an appropriate value for the risk aversion parameter (γ) that closely reflects their risk tolerance. The statistics and distributions of results obtained through simulations that employ stochastic programming models in a rolling horizon context provide insights into the behavior of alternative models, which are valuable for assessment and validation purposes.²

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